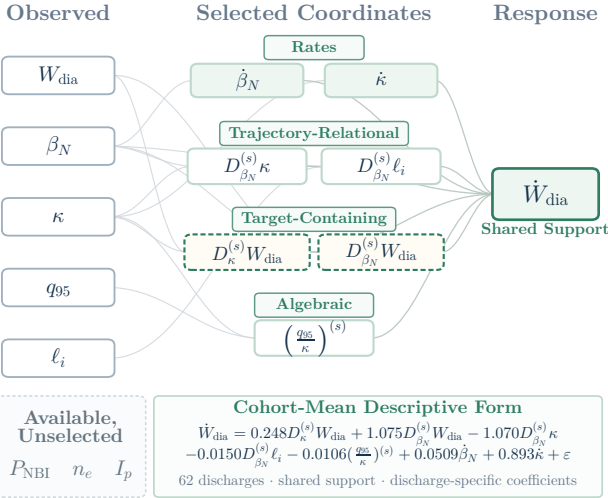


DIII-D: Task-Conditioned Relational Discovery

One 62-discharge observational record $\rightarrow$ Distinct qualified organizations under q_{desc} and q_{rec}

a Descriptive Relational Organization (q_{desc})

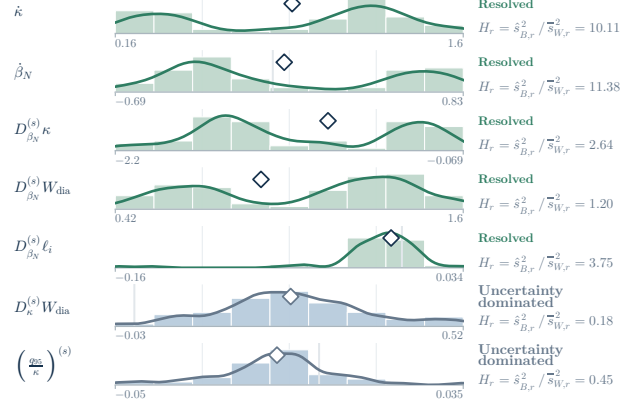


b Coefficient Heterogeneity Audit (q_{desc})

62 discharge-specific coefficients

Decision rule

Resolved when between-discharge variation persists beyond within-discharge uncertainty



Native-scale histogram + KDE; range: observed min-max; diamond: random-effects mean μ_{REML} .

$D_{\beta_N} \ell_i$ is resolved ($H_r = 3.75$, $\tau_{\text{REML}}^2 > 0$);

$D_{\kappa} W_{\text{dia}}$ is uncertainty-dominated ($H_r = 0.18$, $\tau_{\text{REML}}^2 = 0$).

c Recurrent Support Terms Across Folds (q_{rec})

Type	Relational Term	Fold Support						Rec.
		F1	F2	F3	F4	F5	F6	
id	W_{dia}							6
ratio	$\text{ece21} / \text{cerT1i0}$							6
ratio	$n_{\text{c-ped}} / \text{cerT13}$							5
ratio	$t_{\text{inj}} / \text{ece20}$							5
recip	$1 / \text{cerT1i0}$							4
recip	$1 / B_t$							3
ratio	I_p / B_t							3
ratio	$\text{Da03} / \text{cerT13}$							3
prod	$\text{ece37} \times \text{ece39}$							3
prod	$n_{\text{c-ped}} \times \text{cerT16}$							3
ratio	$I_p / \text{ece39}$							3
prod	$P_{\text{inj}} \times \text{ece36}$							2
ratio	$P_{\text{3in}} / \text{cerT1i0}$							2
ratio	$I_p / \text{cerT1i0}$							2
prod	$\text{Do04} \times \text{cerV8}$							2

Ratio
 Product
 Reciprocal
 Identity

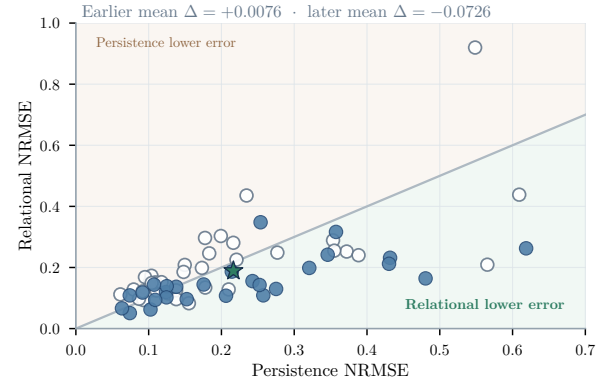
Rows show 15 terms recurring in at least two of six 12-term supports
20 singleton terms are omitted.

Ratio constructions comprise 8 / 15 recurrent terms. Ratio and reciprocal terms shown passed the frozen admissibility and range-support checks in every fold where selected.

d Target-Cross-Fitted Density Reconstruction (q_{rec})

○ Earlier ● Later ★ Mean

mean NRMSE 0.189 vs 0.216
W / T / L 32 / 5 / 25



62 out-of-fold discharges · finite-object, target-cross-fitted qualification
not external validation